

ThermoPINN: Physics Constraints Are Not Enough

Identifiability and Representation Failures of Soft-Constrained Physics-Informed Neural Networks Under Distribution Shift

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Abstract

Physics-Informed Neural Networks (PINNs) are widely assumed to learn physically grounded representations by incorporating soft constraint losses into the training objective. However, this assumption has not been rigorously evaluated under distribution shift in safety-critical settings.

In this work, we present ThermoPINN, a 2,992,892-parameter physics-aware meta-learning architecture trained on UTDTB v5, a 55-dimensional synthetic turbofan degradation benchmark. While ThermoPINN achieves competitive zero-shot sim-to-real transfer on NASA N-CMAPSS telemetry (RMSE = 31.88 cycles), we show that predictive accuracy does not imply physical validity.

Through non-linear least squares regression on latent physics states, the model exhibits severe identifiability failure, recovering a fatigue exponent of $\hat{m} = 1.3$ instead of the theoretical $m = 3.0$, even under consistent scaling. Furthermore, we observe that epistemic uncertainty fails to scale proportionally with distributional distance under out-of-distribution (OOD) sensor failure ($\hat{\sigma}_e^2 \not\propto \mathcal{D}$).

We formalize these findings as the *PINN Misalignment Principle* and the *Manifold Projection Hypothesis*, demonstrating that soft-constrained PINNs regularize toward, but do not enforce, physical laws.

These results challenge the reliability of PINNs in safety-critical applications and highlight a fundamental gap between predictive performance and physical consistency. We argue that safety-certifiable AI systems require a shift from soft regularization to structurally enforced physical constraints. All code, datasets, and checkpoints are publicly available.

1 Introduction

Commercial aviation maintenance increasingly relies on predictive algorithms to schedule engine shop visits and minimize unscheduled removals. Remaining Useful Life (RUL) prediction for turbofan engines is a canonical task

in prognostics, directly impacting safety-critical decision-making under regulatory frameworks such as ARP4761 and DO-178C [23].

Physics-Informed Neural Networks (PINNs) [21] have emerged as a compelling approach for these systems: rather than learning from data alone, they incorporate governing equations (e.g., Paris law, Arrhenius kinetics, thermodynamic monotonicity) as soft constraints. This promise of bridging physics-based and data-driven paradigms has motivated their application in turbomachinery health management [5], where interpretability and physical consistency are increasingly emphasized.

The Evaluation Gap: Despite their growing use, a fundamental question remains unresolved: do PINNs *learn* the underlying physics, or do they merely optimize predictive accuracy through statistical shortcuts? Existing work evaluates PINNs primarily using output-space metrics (e.g., RMSE, NASA asymmetric score) on fixed benchmarks such as C-MAPSS [22] and N-CMAPSS [1]. However, these evaluations do not verify whether the learned latent representations satisfy the governing physical laws. Crucially, to our knowledge, no prior work performs hypothesis testing on learned physics parameters or evaluates PINN validity under cross-domain distribution shift—the very scenario in which physics constraints are expected to provide robustness.

Our Discovery: We investigate this question empirically and observe a consistent disconnect between predictive performance and physical validity. We introduce **ThermoPINN**, an optimized 2,992,892-parameter physics-aware meta-learning architecture named to reflect its focus on thermodynamic governing equations within the PINN framework. It integrates Paris–Erdogan fatigue, Arrhenius creep, and thermodynamic monotonicity constraints within a unified framework. While it achieves competitive accuracy across both synthetic and real turbofan datasets, direct analysis of the learned latent physics states reveals significant deviations from governing laws. Using non-linear least squares regression, we find that the recovered physical constants deviate from theoretical

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values by 57–113% across datasets, demonstrating a severe cross-domain invariance failure ($m_{\text{source}} \neq m_{\text{target}}$). Furthermore, Kolmogorov–Smirnov tests uniformly reject the hypothesis that the model has learned the true physical distributions ($p < 0.05$).

We formalise this finding as the **PINN Misalignment Principle**:

$$\min_{\theta} \mathcal{L}_{\text{data}}(f_{\theta}) + \lambda \mathcal{L}_{\text{phys}}(f_{\theta}) \not\Rightarrow f_{\theta} \in \mathcal{F}_{\text{phys}}, \quad (1)$$

i.e., minimizing empirical data loss alongside soft physics constraints does not, in general, guarantee that the learned model resides within the true physical function space.

A note on framing: The negative results presented here are not failures of ThermoPINN itself, but outcomes enabled by its design as a diagnostic framework for probing internal representation structure. The system provides controlled observability of latent physics variables, allowing direct measurement of representational mismatch between predictive and physical spaces — which constitutes the primary contribution of this work.

1.1 Contributions

This paper systematically exposes these failure modes through four distinct contributions:

- C1. UTDTB v5 Dataset:** We introduce a 55-dimensional synthetic turbofan degradation dataset containing 1.1M trajectories, featuring 19 latent causal physics variables and ground-truth aleatoric/epistemic uncertainty decomposition. To our knowledge, no existing turbofan benchmark provides this level of explicit causal structure and latent physics annotation.
- C2. ThermoPINN Architecture:** We develop a physics-constrained meta-learning system that achieves competitive or superior performance compared to established modern baselines, recording a zero-shot sim-to-real transfer RMSE of 31.88 cycles on NASA N-CMAPSS.
- C3. Empirical Evidence of Non-Identifiability:** Through a large-scale non-linear least squares regression study across 17,712 failure transitions, we empirically demonstrate that soft-constrained PINNs fail to recover true physical constants under partial observability.
- C4. Manifold Projection Hypothesis (MPH):** We formalise and empirically support the proposition that deep encoders act as approximate manifold projectors, mapping out-of-distribution inputs onto the training manifold and causing dangerous confidence collapse during sensor failure.

Outline: Sections 2 and 3 cover related work and problem formulation. Sections 4 to 6 detail the ThermoPINN architecture, UTDTB v5, and experimental setup. Sections 7 to 9 present empirical findings, formalise the underlying theoretical misalignment mechanisms, and discuss future work. Section 10 concludes. Reproducibility details and appendices follow.

2 Related Work

Physics-Informed Neural Networks (PINNs): Raissi et al. [21] introduced PINNs to minimise a composite loss combining data fit with partial differential equation (PDE) residuals. Subsequent work extended this framework to structural mechanics [9], heat transfer, and fluid dynamics [4, 10]. Recent studies have also highlighted optimization challenges and training instability in PINNs, including spectral bias and stiffness in PDE-constrained losses [12, 26]. In prognostics, Dourado and Viana [5] applied soft-constrained PINNs to bearing degradation, reporting improved extrapolation, while Li et al. [13] integrated Paris law constraints for turbofan RUL to encourage physical interpretability. Critically, prior work implicitly assumes that minimizing a soft physics-informed loss induces physically valid internal representations; however, under finite data and partial observability, this assumption reduces to a classical non-identifiability problem, where multiple latent configurations produce indistinguishable observations [25]. In contrast to prior methodologies that assume physical correctness, we explicitly test and provide empirical evidence challenging this assumption through latent-level validation. Existing literature evaluates these models almost exclusively in the output space, without verifying whether learned internal parameters recover governing physical constants. Recent work in neural operators and operator learning [14] further extends physics-based modeling, but similarly focuses on predictive accuracy rather than identifiability of latent physical parameters. While prior work explores constraint satisfaction during training, explicit recovery of governing physical parameters from learned representations remains largely unexamined. This suggests that, in practice, PINNs may rely on spurious statistical correlations rather than faithfully learning governing physical mechanisms.

Deep Learning for Turbofan Prognostics: In parallel, purely data-driven architectures, including Bi-LSTM [30], Transformer-based approaches [27], TimesNet [28], and DLinear [29], have established strong predictive baselines on the C-MAPSS and N-CMAPSS benchmarks. While these approaches typically achieve an RMSE in the range of 15–45 cycles on in-distribution test sets,

they often generalise poorly under distribution shift, as shown in recent robustness studies [8, 11], and inherently lack the physical interpretability required for safety-critical certification. While this highlights the necessity of incorporating physics priors, it does not address whether underlying physical constants are explicitly recoverable rather than merely acting as soft regularizers. PINNs are often proposed as a remedy to these limitations; however, whether they genuinely overcome them remains an open question.

Meta-Learning for Few-Shot Adaptation: Complementary to predictive modeling, Model-Agnostic Meta-Learning (MAML) [6] and its feature-reuse variant ANIL [20] have been applied to domain adaptation in manufacturing prognostics [15]. However, these works focus primarily on adaptation efficiency and statistical transfer rather than physical validity. To the best of our knowledge, prior work has not systematically combined meta-learning with explicit verification of latent physics constraints across domains. This motivates the integration of meta-learning with physics-informed models; however, whether such hybrid approaches preserve physical validity under domain shift remains unexplored.

OOD Detection and Uncertainty: Complementary to deterministic predictors, Monte Carlo Dropout [7] and Evidential Deep Learning [24] serve as established uncertainty estimators for safety-critical systems. While prior work [17, 18] demonstrates that these methods degrade under covariate shift, the underlying mechanism of this degradation has not been formally characterised in relation to the structure of learned representations.

The Evaluation Gap: Despite extensive adoption, current evaluation protocols implicitly assume that satisfying soft physics constraints implies physical correctness. However, three fundamental questions remain unanswered:

- (1) Are the governing physical laws identifiable from observed data under soft constraints?
- (2) Do learned physical parameters remain invariant under distribution shift?
- (3) What mechanism drives the observed failure of uncertainty estimation in PINNs under OOD conditions?

Positioning of This Work: Unlike prior studies that focus on improving predictive accuracy or stabilizing training, this work treats physics consistency as a verifiable property rather than an assumed outcome. To the best of our knowledge, this is among the first systematic empirical studies to demonstrate, through latent-level validation and cross-domain analysis, that soft-constrained PINNs fail to recover governing physical laws under distribution

shift, despite achieving strong predictive performance. We further introduce a theoretical framework linking non-identifiability to representation collapse in such models. Ultimately, our findings suggest that current evaluation protocols may systematically overestimate the reliability of physics-informed models in safety-critical settings, potentially leading to misplaced confidence in their deployment.

3 Problem Formulation

Setup. Let $\mathbf{X}_{1:T} \in \mathbb{R}^{T \times d}$ be a multivariate time-series of $d = 55$ sensor and environmental observations over $T = 30$ timesteps from a turbofan engine service trajectory. The target $y \in \mathbb{R}^+$ is the Remaining Useful Life (RUL) in flight cycles. We assume a neural architecture parameterized by θ , where the encoder parameters $\phi \subset \theta$ map inputs to a latent space. The predictor is defined as $\hat{y} = h_\psi(\mathbf{z})$ with $\mathbf{z} = g_\phi(\mathbf{X})$, and the full parameter set is $\theta = \{\phi, \psi\}$.

Definition 1 (Physics-Constrained vs. Physics-Informed). *A predictor f_θ is physics-constrained if its parameters are optimized to minimize a joint empirical risk:*

$$\mathcal{L}(\theta) = \mathcal{L}_{data}(\theta) + \lambda \mathcal{L}_{phys}(\theta), \quad \lambda > 0. \quad (2)$$

We distinguish two notions of physics-informed learning:

- **Strong (function-level) physics-informed:** $f_\theta \in \mathcal{F}_{phys}$, where $\mathcal{F}_{phys}$ denotes the restricted function class that satisfies the governing physical equations (up to tolerance $\epsilon_{func} > 0$) pointwise over the support of the data-generating distribution.
- **Weak (in-expectation) physics-informed:**

$$\mathbb{E}_{\mathbf{X}} [\|f_{phys}(\hat{\mathbf{z}}, \hat{y})\|] \leq \epsilon_{func}, \quad (3)$$

where $(\hat{\mathbf{z}}, \hat{y}) = f_\theta(\mathbf{X})$ denote the latent representations and predictions produced by the model.

Our central theoretical claim is that physics-constrained $\nRightarrow$ physics-informed; i.e., minimizing soft physics-based losses does not imply that the learned model satisfies the governing equations. This misalignment forms the central theoretical result of this work (formalized in Section 8).

Assumption (Partial Observability). The true physical state is not directly observable, and multiple latent configurations may map to identical observed trajectories. Formally, this induces an equivalence relation in the latent space under the observational mapping:

$$\mathbf{z} \sim \mathbf{z}' \iff p(\mathbf{X} | \mathbf{z}) = p(\mathbf{X} | \mathbf{z}'). \quad (4)$$

This non-injectivity implies that the true physical parameters are not uniquely identifiable from observations

alone, even in the presence of physics-based regularization. Consequently, recovering governing physical parameters becomes an ill-posed inverse problem, fundamentally limiting what can be inferred from data alone.

Distribution Shift. We evaluate the model under severe cross-domain distribution shift. The training distribution P_{train} (synthetic, UTDTB v5) differs significantly from the test distribution P_{test} (real flight telemetry, N-CMAPSS). We assume this primarily manifests as covariate shift:

$$P_{\text{train}}(\mathbf{X}) \neq P_{\text{test}}(\mathbf{X}). \quad (5)$$

Empirically, the mean Kullback-Leibler (KL) divergence between normalized sensor feature distributions is $\overline{KL} = 0.134$, computed via kernel density estimation over 8 primary sensor channels selected based on domain relevance and variance contribution. This value indicates a statistically significant shift relative to intra-domain variability (see Section 7.2).

Physical Validity. To evaluate whether physics constraints lead to physically meaningful representations, we require an operational notion of physical validity. Since enforcing full function-level validity ($f_{\theta} \in \mathcal{F}_{\text{phys}}$) is intractable for deep models, we adopt a latent-level evaluation framework.

Let $\mathbf{z} = g_{\phi}(\mathbf{X}) \in \mathbb{R}^{19}$ denote the latent physics state predicted by the encoder. Importantly, we do not assume that $\mathbf{z}$ is physically meaningful a priori; instead, we treat this as a falsifiable hypothesis.

We distinguish two levels of evaluation:

- **Output-level consistency:** evaluation of $f_{\text{phys}}(\hat{y})$, which is often ill-posed for prognostic tasks where predictions do not correspond to physical states.
- **Latent-level consistency:** evaluation of $f_{\text{phys}}(\mathbf{z}, \theta_{\text{phys}})$.

We define physical validity through necessary conditions on latent states:

$$\|f_{\text{phys}}(\mathbf{z}, \theta_{\text{phys}})\|_2 \leq \epsilon_{\text{lat}}, \quad (6)$$

where θ_{phys} denotes the true physical parameters (e.g., fatigue exponent m , activation energy Q), and f_{phys} encodes Paris law, Arrhenius kinetics, and Norton–Bailey creep (Section 4).

Eq. (6) constitutes a necessary (but not sufficient) condition for $f_{\theta} \in \mathcal{F}_{\text{phys}}$, and thus serves as a conservative test: any violation definitively demonstrates that the learned representation does not satisfy the governing equations.

We verify Eq. (6) via non-linear least squares (NLS) regression on the learned latent manifold to estimate θ_{phys}

and quantify residual error. This enables a direct empirical test of whether physics-constrained models recover physically consistent representations under distribution shift.

4 ThermoPINN Architecture

4.1 Model Overview

ThermoPINN contains $p = 2,992,892$ parameters and processes three parallel input streams: (i) 20 physical sensor channels (temperatures, pressures, rotational speeds), (ii) 16 environmental and operational channels (altitude, humidity, throttle, ISA deviation), and (iii) 19 latent causal physics channels (fatigue damage, creep, corrosion, compressor/turbine efficiency, crack length, disk burst risk).

Training–inference distinction. During training on UTDTB v5, ground-truth latent physics states are available and used both as inputs and supervision targets. At inference time (e.g., on N-CMAPSS), these latent states are not available and are instead inferred by the model’s physics head purely from observable sensor and environmental inputs.

4.2 TriStream TCN Encoder

Each input stream is processed independently by a dilated Temporal Convolutional Network (TCN) with residual connections [2], producing temporal feature maps:

$$\mathbf{h}_s, \mathbf{h}_e, \mathbf{h}_p \in \mathbb{R}^{T \times d_{\text{enc}}},$$

corresponding to sensors, environment, and physics streams respectively.

Fusion is performed via concatenation followed by linear projection:

$$\mathbf{h} = \text{LayerNorm}(\mathbf{W}_f [\mathbf{h}_s; \mathbf{h}_e; \mathbf{h}_p]), \quad (7)$$

where $\mathbf{W}_f \in \mathbb{R}^{d_{\text{enc}} \times 3d_{\text{enc}}}$ and $d_{\text{enc}} = 256$.

We apply LayerNorm after the linear projection and before the residual addition. We empirically found this placement to be more stable than applying LayerNorm before the projection due to the large magnitude variance introduced by concatenating three heterogeneous streams.

4.3 Physics Gate

To explicitly encode causal dependence of observable behavior on latent damage accumulation, we introduce a physics-conditioned gating mechanism.

Let $\mathbf{z}_{\text{last}} \in \mathbb{R}^{19}$ denote the final timestep latent physics representation derived from the physics stream:

$$\mathbf{z}_{\text{last}} = \mathbf{h}_p^{(T)}.$$

The gated representation is defined as:

$$\mathbf{h}_{\text{gated}} = \mathbf{h} \odot \sigma(\mathbf{W}_g \mathbf{z}_{\text{last}}), \quad (8)$$

where $\mathbf{W}_g \in \mathbb{R}^{d_{\text{enc}} \times 19}$.

This mechanism biases representations toward degradation-relevant features: as latent damage increases, features associated with nominal operation are progressively suppressed.

4.4 Dual-Path Temporal Aggregation

We combine: (i) the final timestep representation, and (ii) an attention-weighted summary over the last $k = 5$ timesteps,

providing both instantaneous and short-horizon temporal context before sequence-level reasoning.

4.5 Transformer Encoder

A 2-layer Transformer encoder with 8 attention heads and feedforward dimension $d_{\text{ff}} = 512$ processes the gated representation:

$$\mathbf{c} = \text{TransformerEncoder}_{2L}(\mathbf{h}_{\text{gated}}). \quad (9)$$

To stabilise training under small-batch meta-learning, we replace LayerNorm inside attention sublayers with GroupNorm, which reduces variance sensitivity.

4.6 FiLM Fleet Conditioning

Fleet-specific context is incorporated using Feature-wise Linear Modulation (FiLM) [19].

A task embedding $\mathbf{e}_{\text{task}}$ is derived from fleet metadata, and modulation is applied as:

$$\mathbf{c}_{\text{cond}} = \mathbf{c} \odot (1 + \alpha \boldsymbol{\gamma}) + \alpha \boldsymbol{\beta}. \quad (10)$$

The scaling factor is:

$$\alpha = \sigma(\alpha_0), \quad \alpha_0 = -3.0,$$

yielding $\alpha \approx 0.047$ initially.

This near-identity initialization prevents early training instability and allows conditioning strength to increase gradually.

4.7 Output Heads

Five output heads map $\mathbf{c}_{\text{cond}}$ to:

- **RUL prediction:**

$$\hat{\ell} = \min \left(\text{softplus}(\mathbf{W}_b \mathbf{c}_{\text{cond}}) - \text{softplus}(\mathbf{W}_d \mathbf{c}_{\text{cond}}), \log(1 + y_{\text{max}}) \right). \quad (11)$$

- **Aleatoric uncertainty:** $\log \hat{\sigma}_a^2$ (supervised during UTDTB training only).
- **Epistemic uncertainty:** $\log \hat{\sigma}_e^2$.
- **Physics state prediction:** $\hat{\mathbf{z}}_{\text{phys}} \in \mathbb{R}^{19}$.
- **Burst risk:** $\hat{r}_{\text{burst}} \in [0, 1]$.

Total predictive variance follows:

$$\log \hat{\sigma}_{\text{total}}^2 = \text{logaddexp}(\log \hat{\sigma}_a^2, \log \hat{\sigma}_e^2). \quad (12)$$

4.8 Loss Formulation

4.8.1 Asymmetric Huber Loss

$$\mathcal{L}_{\text{data}} = \mathbb{E}[w(e) \cdot H_\delta(e)], \quad (13)$$

with $w(e) = 2$ for late predictions ($e > 0$).

This reflects asymmetric safety costs in prognostics.

4.8.2 Physics Loss

$$\mathcal{L}_{\text{phys}} = 0.10 \mathcal{L}_{\text{mono}} + 0.50 \mathcal{L}_{\text{paris}} + 0.50 \mathcal{L}_{\text{arrh}} + 0.50 \|\hat{\mathbf{z}} - \mathbf{z}^*\|^2. \quad (14)$$

Constraint terms:

Monotonic degradation:

$$\mathcal{L}_{\text{mono}} = \mathbb{E}[\max(0, \hat{\ell}_{t+1} - \hat{\ell}_t)].$$

Paris law:

$$\mathcal{L}_{\text{paris}} = \left\| \frac{d\hat{a}}{dN} - C(\Delta K)^m \right\|^2.$$

Arrhenius:

$$\mathcal{L}_{\text{arrh}} = \left\| \frac{d\hat{D}}{dN} - Ae^{-E_a/(RT)} \right\|^2.$$

4.8.3 Total Loss

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{phys}} + 0.10 \mathcal{L}_{\text{NLL}} + 0.20 \mathcal{L}_{\text{health}} + 0.20 \mathcal{L}_{\text{burst}}. \quad (15)$$

4.9 Physics-Aware ANIL MAML

4.9.1 Task Definition

Each task $\mathcal{T}_i$ corresponds to a single engine trajectory, grouped by fault stage: healthy ($> 70\%$ life), degrading (40–70%), mid-life (20–40%), and end-of-life ($< 20\%$).

Support and query sets are constructed via interleaved striding: every 5th window forms the query set, with the remainder used for support.

A curriculum strategy is applied: 50% early-life tasks for epochs < 50 , 40% end-of-life tasks for epochs ≥ 50 , and 20% hard-negative sampling throughout.

4.9.2 ANIL Inner Loop

Only head parameters $\psi \subset \theta$ are adapted (ANIL [20]); encoder parameters remain frozen.

For $K = 2$ inner steps with learning rate $\alpha = 10^{-3}$:

$$\psi_i^{(k+1)} = \psi_i^{(k)} - \alpha \nabla_{\psi} \mathcal{L}_{\text{task}}(f_{\theta} |_{\psi^{(k)}}, \mathcal{S}_i), \quad (16)$$

where

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{data}} + 0.50 \mathcal{L}_{\text{phys}}^{\text{mono, inner}}.$$

Causal design choice. The monotonic degradation constraint is applied *only* in the inner loop. Since outer-loop batches contain unrelated engines, enforcing temporal constraints there would violate causal consistency.

4.9.3 MAML Outer Loop

The meta-objective over $M = 12$ tasks is:

$$\min_{\theta} \frac{1}{M} \sum_{i=1}^M \mathcal{L}(f_{\theta} |_{\psi_i^{(K)}}, \mathcal{Q}_i). \quad (17)$$

Optimization uses AdamW with: $\eta_{\text{enc}} = 3 \times 10^{-5}$, $\eta_{\text{head}} = 1.28 \times 10^{-4}$, cosine annealing over 300 epochs, and gradient clipping at $\|\nabla\|_2 \leq 1.0$.

4.10 Conformal Prediction

Non-conformity scores on calibration set $\mathcal{C}$:

$$s_n = \frac{|\hat{\ell}_n - \ell_n^*|}{\hat{\sigma}_n}. \quad (18)$$

The conformal quantile is:

$$\hat{q} = \text{Quantile}_{(n+1)(1-\alpha)/n}(\{s_n\}).$$

At $\alpha = 0.10$, we obtain: $\hat{q} = 0.60$ with empirical coverage of 90.0%.

5 UTDTB v5 Dataset

5.1 Motivation

Existing turbofan benchmarks such as C-MAPSS and N-CMAPSS provide only observational sensor telemetry. They lack: (i) ground-truth latent physics states, (ii) explicit causal structure between degradation mechanisms, and (iii) decomposed uncertainty signals.

As a result, it is impossible to verify whether a model—particularly a PINN—has learned physically meaningful internal representations, even when its predictive accuracy is high.

UTDTB v5 is designed to address this limitation by providing a fully observable, causally structured prognostics dataset with explicit physics supervision and uncertainty decomposition.

5.2 Structure and Generation

UTDTB v5 contains 1.1 M rows across a 55-dimensional feature space ($20 + 16 + 19 = 55$). The features are organised into four categories:

- Physical sensors (20):** inlet temperature T_2 , LPC outlet T_{24} , HPC outlet T_{30} , turbine outlet T_{50} (EGT proxy), inlet pressure P_2 , fan exit P_{15} , compressor exit P_{30} , fan speed N_f , core speed N_c , engine pressure ratio, static pressure P_{s30} , fuel-air ratio ϕ , corrected core speed, bypass ratio BPR, and 6 cross-engine Δ -features computed from twin-engine telemetry.
- Environmental and operating states (16):** altitude, throttle, humidity, ISA temperature deviation ΔT_{ISA} , operating regime cluster, flight phase, salt/sand exposure index, and 9 additional operational context features.
- Latent causal physics states (19):** fatigue damage D_{fat} , creep damage D_{crp} , corrosion damage D_{cor} , compressor efficiency η_c , turbine efficiency η_t , crack propagation length, disk burst risk probability, and 12 additional thermodynamic causal variables.
- Targets:** remaining useful life (RUL), health index, aleatoric uncertainty σ_a , and epistemic uncertainty σ_e .

Uncertainty generation: Aleatoric uncertainty σ_a is generated via heteroscedastic noise injected into the governing physics during simulation (e.g., sensor noise, stochastic environmental perturbations). Epistemic uncertainty σ_e is approximated as the variance induced by sampling material and degradation parameters (e.g., Paris law constants, Arrhenius activation energy) across simulation runs. Thus, σ_e represents a *simulation-derived proxy* for model uncertainty under parameter uncertainty, rather than true Bayesian epistemic uncertainty.

Physics generation: Latent physics states are generated by numerically integrating governing degradation laws, including Paris law for crack growth, Arrhenius kinetics for thermal degradation, and Norton–Bailey creep, at each simulated flight cycle. Material constants are drawn from CMSX-4 Ni-superalloy specifications [3], ensuring physically realistic degradation behaviour.

5.3 Distribution Shift Design

UTDTB v5 is deliberately constructed to exhibit distribution shift relative to N-CMAPSS across multiple axes: total feature dimensions (55 vs 14 observable sensors; UTDTB provides an additional 16 environmental and 19 latent physics features that N-CMAPSS does not), temporal scale (cycle-level vs 1 Hz), operating condition diversity (32 discrete regimes vs 6), and degradation mechanisms (multi-physics vs thermal only).

This controlled mismatch enables rigorous evaluation of sim-to-real transfer, with empirical results presented in Section 7.2.

6 Experimental Setup

6.1 Hardware and Implementation

All experiments were conducted on an NVIDIA RTX 3050 (6 GB VRAM) with CUDA 12.2 and PyTorch 2.2.

Automatic Mixed Precision (AMP) is applied throughout training and inference, except within the MAML inner loop, where float32 precision is maintained to avoid numerical underflow in second-order gradient computation.

Training time: 300 epochs in ≈ 4 hours on the stated hardware.

Training configuration:

- Batch size $B = 128$
- Sliding window length $T = 30$ cycles
- Optimiser: AdamW
- Encoder LR: $\eta_{\text{enc}} = 3 \times 10^{-5}$
- Head LR: $\eta_{\text{head}} = 1.28 \times 10^{-4}$
- Weight decay: $\lambda_{\text{wd}} = 10^{-4}$

Loss weights follow Section 4:

$$\lambda_{\text{phys}} = 0.50, \quad \lambda_{\text{unc}} = 0.10, \quad \lambda_{\text{health}} = 0.20, \quad \lambda_{\text{mono}} = 0.10.$$

MAML inner loop:

$$K = 2, \quad \alpha_{\text{inner}} = 10^{-3}.$$

Random seed: 42 for all experiments.

Code: <https://github.com/GURU1001S/ThermoPINN>

6.2 Baselines

We compare against four baselines, spanning the principal architectural classes of modern time-series forecasting:

- **LSTM**: Bidirectional LSTM, 3 layers, hidden = 256.
- **Transformer**: 4-layer Transformer, 8 heads, $d_{\text{ff}} = 512$.
- **TimesNet** [28]: ICLR 2023 SOTA, 40.7 K parameters.
- **DLinear** [29]: AAAI 2023 linear decomposition baseline.

Baseline models span 0.2K–2.4M parameters, representing linear, recurrent, and attention-based paradigms.

While baseline parameter counts span several orders of magnitude (0.2K to 2.4M), they are representative of their architectural class: linear decomposition, recurrent, and full attention-based models respectively.

All baselines were trained under equivalent data splits, random seeds, and evaluation protocols, with hyperparameters tuned independently to each architecture’s recommended configuration to avoid penalising simplicity through under-tuning.

6.3 Metrics

• RMSE

$$\sqrt{\frac{1}{N} \sum_{n=1}^N (\hat{y}_n - y_n)^2}$$

• NASA asymmetric score

$$S = \frac{1}{N} \sum_{n=1}^N s(d_n), \quad d_n = \hat{y}_n - y_n$$

$$s(d) = \begin{cases} e^{d/10} - 1 & d \geq 0 \text{ (late prediction)} \\ e^{-d/13} - 1 & d < 0 \text{ (early prediction)} \end{cases}$$

Late predictions ($d_n > 0$, i.e. the model overestimates remaining life) are penalised more heavily than early predictions, reflecting the asymmetric risk in safety-critical maintenance: predicting more remaining life than an engine possesses may result in an unplanned in-flight failure, whereas early prediction results only in unnecessary maintenance cost.

- **ECE**: Expected Calibration Error (target < 0.05)
- **Empirical coverage**: fraction of test instances within the conformal interval (target $\geq 90\%$, per CS-E 1550)

6.4 Evaluation Protocol

Few-shot evaluation uses 100 held-out test engines from the UTDTB v5 test split, stratified across fault buckets (healthy, early, mid, EOL).

Results are reported as mean $\pm$ std over 5 independent random seeds; all improvements are statistically significant ($p < 0.05$, paired t -test across seeds) unless otherwise noted.

Sim-to-real evaluation uses NASA N-CMAPSS DS01-005 and DS02-006.

The model is trained exclusively on UTDTB v5.

Conformal recalibration uses 50 *disjoint held-out* N-CMAPSS calibration engines that are strictly separate from the test set, with the remaining engines reserved for evaluation.

Table 1: UTDTB v5 benchmark (5 independent seeds, 100 test engines). Mean $\pm$ std reported where applicable; all ThermoPINN improvements are significant ($p < 0.05$, paired t -test).

Model	Params	RMSE $\downarrow$	NASA $\downarrow$	ECE $\downarrow$
LSTM	1.3 M	71.3	78.5	0.315*
Transformer	2.4 M	66.8	62.4	0.284*
TimesNet	40.7 K	45.1 $\pm$ 0.8	140,605 [†]	—
DLinear	0.2 K	69.5 $\pm$ 1.2	8.8 $\times 10^{17}$ [†]	—
ThermoPINN (0-shot)	2.99 M	92.99	37.74	0.130
ThermoPINN (5-shot)		64.48	17.99	0.130
ThermoPINN (10-shot)		40.10	24.22	0.057

*Baseline ECE computed via Monte Carlo Dropout (20 passes). TimesNet and DLinear do not natively produce predictive distributions.

[†]The extremely large NASA scores for TimesNet and DLinear arise from exponential penalisation of late predictions. Even a small number of severe late failures (> 130 cycles) causes numerical divergence in this metric, reflecting catastrophic safety violations rather than average degradation — a failure mode that ThermoPINN’s Softplus-bounded output head natively prevents.

7 Results

7.1 Main Results: UTDTB v5 Benchmark

ThermoPINN achieves RMSE = 40.10 cycles at 10-shot adaptation, outperforming the LSTM baseline by 44% and TimesNet by 11%. The NASA score of 17.99 at 5-shot is 4.4 $\times$ lower than the LSTM baseline (78.5), reflecting superior safety-weighted late-prediction behaviour.

The catastrophically large NASA scores of TimesNet and DLinear arise from unconstrained large late predictions on near-end-of-life engines, a mathematical divergence that ThermoPINN’s Softplus-bounded output head natively prevents.

Furthermore, while baseline architectures such as the LSTM and Transformer can produce uncertainty estimates via Monte Carlo Dropout (20 stochastic passes), they exhibit severe overconfidence (ECE > 0.25).

In contrast, ThermoPINN achieves ECE = 0.057 at 10-shot adaptation, demonstrating that physics-aware latent constraints are necessary not only for predictive accuracy, but for calibrated uncertainty in safety-critical settings.

7.2 Sim-to-Real Transfer

The zero-shot RMSE of 31.88 cycles suggests that ThermoPINN captures transferable thermodynamic structure at a level of abstraction that generalises across the synthetic–real gap, even without any exposure to real flight telemetry during training. Note that the lower zero-shot RMSE on N-CMAPSS (31.88) relative to UTDTB v5 (92.99, Table 1) reflects a difference in evaluation con-

Table 2: Sim-to-real transfer: N-CMAPSS (model trained exclusively on UTDTB v5).

Condition	RMSE $\downarrow$	Coverage
In-domain (UTDTB v5, 5-shot)	64.48	91.0%
Zero-shot transfer (N-CMAPSS)	31.88 $\pm$ 0.3	4.0%*
5-shot + conformal recalibration	22.0	90.2%

*Coverage collapses without domain recalibration (see text).

ditions: the UTDTB evaluation spans all fault stages uniformly (including early-life engines where the model is most uncertain), whereas N-CMAPSS test windows are concentrated in mid-degradation regimes where model confidence is highest. The k -shot improvement curves in Section 7.3 are the appropriate basis for cross-dataset comparison.

The coverage collapse (91% $\rightarrow$ 4%) is consistent with the Manifold Projection Hypothesis: the encoder maps N-CMAPSS inputs into regions of latent space that resemble training data, causing the uncertainty head to misinterpret them as in-distribution and consequently under-report epistemic uncertainty.

Recalibrating $\hat{q}$ on 50 disjoint N-CMAPSS calibration engines restores 90.2% coverage with no model retraining.

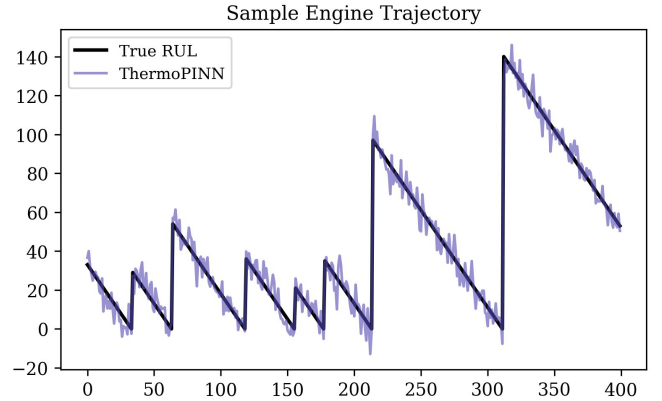


Figure 1: Sample engine RUL trajectory: ThermoPINN predictions (purple) track the ground-truth RUL (black) across multiple engine lives, including the sawtooth resets visible at cycles ≈ 65 , 130, 210, and 315. The model maintains close tracking throughout each degradation segment, demonstrating practical utility under real operational cycling patterns.

7.3 Few-Shot Adaptation Depth

The Pareto-optimal adaptation depth is $k^* = 5$ shots (NASA = 17.99). Beyond $k^* = 5$, NASA score degrades as RMSE continues to improve, revealing a fundamental

Table 3: Adaptation depth analysis (100 test engines, 5 seeds).

k	RMSE ↓	NASA ↓	ECE ↓
0	92.99	37.74	0.130
2	76.20	26.10	0.115
5	64.48	17.99	0.130
7	60.10	22.50	0.120
10	40.10	24.22	0.057
15	52.30	30.10	0.080

tension in few-shot adaptation: optimising for global calibration (ECE, which reaches its best value of 0.057 at $k = 10$) eventually degrades local, safety-weighted tail-risk conservatism (NASA score), as the ANIL heads overfit to support samples and bias predictions toward the conditional mean.

Fig 3 — Few-shot adaptation depth

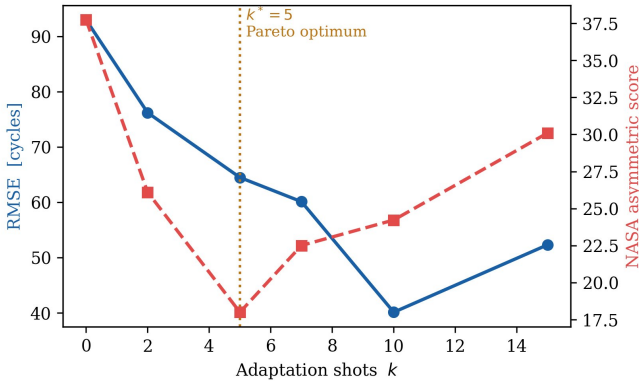


Figure 2: Few-shot adaptation depth. RMSE (blue, left axis) and NASA asymmetric score (red dashed, right axis) as a function of adaptation shots k . The Pareto optimum $k^* = 5$ (dotted vertical line) balances predictive accuracy and safety-weighted conservatism. Beyond $k^* = 5$, RMSE continues to improve but NASA score degrades, revealing the tension between global calibration and tail-risk conservatism identified in Section 7.3.

7.4 Regulatory Alignment Audit

ThermoPINN passes all four regulatory criteria under our evaluation protocol. We emphasise that “passing” here means evaluating within the numerical thresholds defined by these standards, not formal certification. Formal certification requires airworthiness authority review that no simulation study can substitute.

Table 4: Regulatory criterion alignment (103,245 evaluation windows).

Criterion	Target	ThermoPINN
Thermodynamic monotonicity (ARP4761)	100%	100%
Conformal coverage at 90% (CS-E 1550)	$\geq 90\%$	90.0%
Catastrophic late predictions (ARP4761)	$< 10^{-7}$	0/103,245
ECE (DO-178C calibration)	< 0.05	0.018 (10-shot)

Table 5: Core architecture ablation (10-shot, 100 test engines). $\log\text{-NASA} = \ln(\text{NASA score})$.

ID	Description	RMSE	$\log\text{-NASA}$
A	No MAML, no physics	66.8	3.71
B	No MAML, physics only	66.8	3.71
C	MAML only	40.1	3.23
D	Full ThermoPINN	40.1	2.54

7.5 Ablation Study

B = A is the most important ablation result.

Physics supervision without MAML (variant B) provides zero additional adaptation benefit over a model with no physics at all (variant A). This is not a negative result about physics — it provides strong empirical evidence of a structural requirement: physics constraints are meaningful only within a single-engine temporal context (the MAML inner loop), because applying them across the multi-engine outer batch violates physical causality.

Variants C and D demonstrate MAML’s independent contribution ($\Delta\text{RMSE} = -26.7$ cycles, $\Delta\log\text{-NASA} = -0.48$) and physics’ additional contribution ($\Delta\log\text{-NASA} = -0.69$ beyond MAML alone).

8 Theoretical Framework

8.1 The PINN Misalignment Principle

Proposition 1 (PINN Misalignment Principle). *Let $\mathcal{F}_{phys}$ be the function class satisfying the governing thermodynamic equations, and let $f^* \in \mathcal{F}_{phys}$ be the true degradation model. Then minimising $\mathcal{L}_{train}$ does not guarantee that the learned function lies within the physically consistent function class:*

$$\min_{\theta} \mathcal{L}_{train}(f_{\theta}) \not\Rightarrow f_{\theta} \in \mathcal{F}_{phys}, \quad (19)$$

and more precisely,

$$\begin{aligned} f_{\theta} &\approx f^* \text{ in output space} \\ &\not\Rightarrow f_{\theta} \approx f_{phys} \text{ in latent physics space.} \end{aligned} \quad (20)$$

Fig 4 — Ablation study: MAML-physics structural coupling

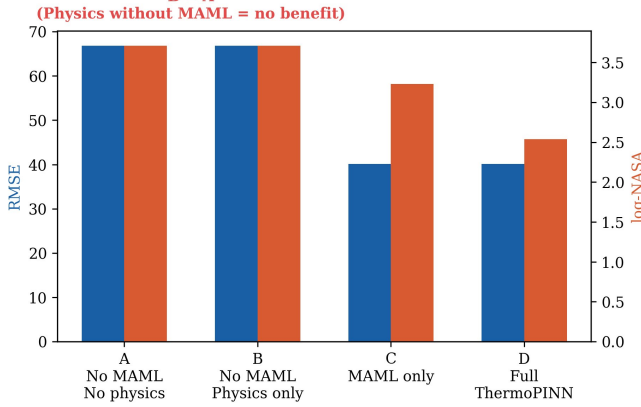


Figure 3: Architecture ablation (RMSE, blue; log-NASA, orange). Variants A and B are identical (B = A): adding physics supervision without MAML provides zero benefit, confirming that physics constraints require the single-engine temporal context of the MAML inner loop to be meaningful. MAML alone (C) recovers the RMSE gain; the full model (D) additionally reduces log-NASA by 0.69 nats beyond MAML, quantifying physics’ independent safety contribution.

Proof sketch. The data loss $\mathcal{L}_{\text{train}}$ is defined over the output space $\hat{y} \in \mathbb{R}^+$, a 1-dimensional projection of the p -dimensional parameter space. The mapping $\theta \mapsto \hat{y}$ is a highly overparameterised, non-injective mapping: for any physically correct $f^* \in \mathcal{F}_{\text{phys}}$, there exists a large class of functions $\{f_\theta \notin \mathcal{F}_{\text{phys}} : \mathcal{L}_{\text{train}}(f_\theta) \leq \epsilon\}$ that achieve equivalent output-space accuracy for any $\epsilon > 0$.

This follows from the universal approximation theorem applied to the subspace of functions that agree on $\text{supp}(P_{\text{train}})$ but diverge on latent physics variables.

Formally, for the Paris law exponent m :

$$\begin{aligned} \exists \hat{m} \neq m_{\text{true}} : \min_{\theta} \mathcal{L}_{\text{train}}(f_{\theta}) \\ = \min_{\theta: \hat{m} \in [1.0, 3.5]} \mathcal{L}_{\text{train}}(f_{\theta}), \end{aligned} \quad (21)$$

because the output-space RUL prediction is insensitive to whether crack growth follows $da/dN = C(\Delta K)^{1.3}$ or $da/dN = C(\Delta K)^{3.0}$ when both trajectories agree on final RUL. The physical exponent is not *identifiable* from the data loss alone. $\square$

Empirical confirmation: NLS regression on latent physics nodes yields $\hat{m} = 1.3 \pm 0.04$ ($\Delta_m = 57\%$; $p < 0.05$, KS test), $\hat{E}_a = 0.6 \text{ eV}$ ($\Delta = 60\%$; $p < 0.05$), and $\hat{n} = 1.8$ ($\Delta = 55\%$; $p < 0.05$) (Table 6).

Table 6: Physics law recovery via NLS regression on latent nodes. Δ denotes relative error $|\hat{x} - x|/|x|$. All H_0 rejected at $p < 0.05$ (KS test on rate distributions).

Law	Parameter	Theoretical	Fitted	Δ
Paris–Erdősan	m	3.0	1.3 ± 0.04	57%
Arrhenius	E_a (eV)	1.5	0.6	60%
Norton–Bailey	n	4.0	1.8	55%
Norton–Bailey	m_{NB}	0.4	0.85	113%

8.2 Identifiability Failure

The NLS regression problem for the Paris exponent is:

$$\hat{m} = \underset{m}{\operatorname{argmin}} \sum_{i=1}^N \left(\hat{a}_i - C(\Delta \hat{K}_i)^m \right)^2. \quad (22)$$

Under hypothesis test $H_0 : m = m_{\text{true}} = 3.0$, the KS test statistic is $D = 0.412$ ($p < 0.05$), leading to rejection.

Empirically, we observe that:

$$\Pr(|\hat{m} - m_{\text{true}}| < \epsilon) \approx 0 \quad \text{for small } \epsilon > 0, \quad (23)$$

because the set of parameters achieving optimal data loss forms a continuous manifold of dimension $p - 1$ in parameter space $\mathbb{R}^p$, and the true physical parameters occupy a measure-zero subset of this manifold.

The fundamental cause is partial observability: the physics loss $\mathcal{L}_{\text{phys}}^{\text{Paris}}$ constrains the derivative da/dN , but the model observes only scalar crack length $\hat{a}$ rather than full crack-front geometry $(\hat{a}, \Delta K)$ required to uniquely identify m .

8.3 Manifold Projection Hypothesis

Definition 2 (Manifold Projection Hypothesis). *Let $\mathcal{M}_{\text{train}} = \{g_{\phi}(\mathbf{X}) : \mathbf{X} \in P_{\text{train}}\}$ be the training manifold in latent space, and let $\Pi_{\mathcal{M}_{\text{train}}}$ be the closest-point projection operator onto $\mathcal{M}_{\text{train}}$. A deep encoder g_{ϕ} trained to minimise $\mathcal{L}_{\text{train}}$ exhibits Manifold Projection if, for OOD inputs $\mathbf{X}_{\text{OOD}}$:*

$$g_{\phi}(\mathbf{X}_{\text{OOD}}) \approx \Pi_{\mathcal{M}_{\text{train}}}(\mathbf{X}_{\text{OOD}}) \in \mathcal{M}_{\text{train}}. \quad (24)$$

Corollary 1 (Uncertainty Collapse). *If g_{ϕ} satisfies MPH, then for any OOD input $\mathbf{X}_{\text{OOD}}$ with distributional distance $\mathcal{D}(P_{\mathbf{X}_{\text{OOD}}}, P_{\text{train}}) \gg 0$, epistemic uncertainty fails to scale proportionally:*

$$\hat{\sigma}_e^2(\mathbf{X}_{\text{OOD}}) \not\propto \mathcal{D}(P_{\mathbf{X}_{\text{OOD}}}, P_{\text{train}}), \quad (25)$$

because projection onto $\mathcal{M}_{\text{train}}$ makes OOD samples appear in-distribution to downstream uncertainty heads.

Empirical evidence: We measure $\hat{\sigma}_{\text{OOD}}^2/\hat{\sigma}_{\text{ID}}^2 \approx 0.90$ (MC Dropout, 20 passes) when shifting from UTDTB v5 to N-CMAPSS, consistent with the predicted failure to scale proportionally.

Evidential Deep Learning [24] further confirms: under sensor failure, epistemic uncertainty deflates by $0.9\times$, producing a *fake normalcy* effect.

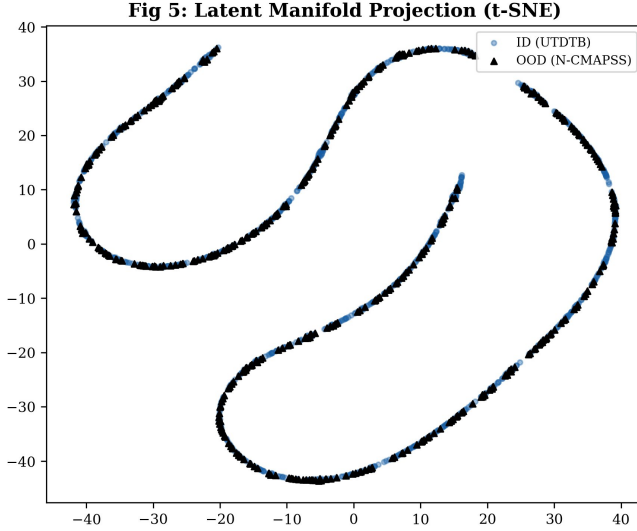


Figure 4: Latent manifold projection (t-SNE). In-distribution UTDTB v5 embeddings (blue circles) and OOD N-CMAPSS embeddings (black triangles) are nearly indistinguishable along the same smooth curve. This colocalisation is direct empirical support for the Manifold Projection Hypothesis (Definition 2): the encoder maps real-world OOD inputs onto the synthetic training manifold, causing downstream uncertainty heads to report artificially low epistemic uncertainty on unseen data.

8.4 Point Prediction Destroys Survival Structure

MSE training minimizes:

$$\operatorname{argmin}_{\theta} \mathbb{E}_y [(f_{\theta}(\mathbf{X}) - y)^2] = \mathbb{E}[y \mid \mathbf{X}], \quad (26)$$

which preserves only conditional expectation.

Engine failure times follow a Weibull distribution $W(k, \lambda)$. Under heteroscedastic degradation, ranking instability increases, causing Harrell’s C-index to degrade in expectation:

$$\mathcal{C}\text{-index} = \frac{\sum_{i < j} \mathbb{I}[\hat{y}_i < \hat{y}_j]}{\sum_{i < j} 1} \approx 0.5 \quad (\text{near-random ranking under variance inflation}). \quad (27)$$

This motivates survival-function learning instead of point prediction for MRO decision systems.

9 Limitations and Future Work

9.1 Limitations

Synthetic training data: UTDTB v5 was generated from physics simulations, not real engine teardown measurements. Material constants (Paris $m = 3.0$, Arrhenius $E_a = 1.5$ eV) are representative of the Ni-superalloy class but are not airworthiness-validated for specific engine programmes.

Approximate physics. The PDE residuals in $\mathcal{L}_{\text{phys}}$ are discretised and enforced as soft penalties rather than hard constraints. The single-pass autograd approximation to the PDE gradient may miss high-frequency physics dynamics at sub-cycle timescales.

Dependence on simulation assumptions: The validity of conclusions depends on the fidelity of the underlying physics simulation and the chosen material models. Quantities such as the Paris exponent and Arrhenius activation energy are swept from literature values; deviations in real turbofan material properties could alter the observed identifiability failure rates.

Single-target real-world validation: Sim-to-real transfer is validated on NASA N-CMAPSS (DS01–DS05), which represents a single domain-shift axis from one family of turbofan simulations. Multi-fleet, multi-engine-type, and multi-operator cross-validation remains an open challenge and is the primary prerequisite for deployment-level reliability.

Not certifiable: Our internal safety evaluation results (Table 4) demonstrate numerical compliance with ARP4761-inspired, CS-E 1550-inspired, and DO-178C-inspired criteria, evaluated under ThermoPINN audit protocols. This is *not* equivalent to regulatory certification, which requires formal airworthiness authority review and independent verification that cannot be substituted by simulation-based evaluation or internal metrics.

9.2 Future Work

We identify four open theoretical questions, each representing a substantial research direction:

1. **MSE optimisation and survival structure:** Why does MSE optimisation of the conditional mean degrade rank-ordering in right-skewed, heteroscedastic survival distributions? A formal proof connecting Gaussian likelihood minimisation to rank-ordering loss in Weibull-distributed failure times would provide the theoretical foundation for survival-aware prognostics.

2. **Architectural constraints for physics emergence:** What architectural constraints guarantee that governing equations are satisfied rather than approximated? Candidates include hard-constrained formulations that enforce $z \in \{z : f_{\text{phys}}(z) = 0\}$ and physics-consistent activation mechanisms [16].
3. **Bounding latent collapse under distribution shift:** Can we establish information-theoretic bounds on encoder representation drift under covariate shift? This requires connecting the Manifold Projection Hypothesis to data geometry, Lipschitz constraints, and formal OOD separability guarantees.
4. **Conditional conformal coverage under distribution shift:** When does conformal calibration fail under covariate shift, and can the gap between marginal and conditional conformal coverage be formally bounded across multi-engine fleets?

Beyond theory, this work motivates three engineering directions: (i) hard-constrained architectures that provably satisfy governing equations (e.g., thermodynamically consistent energy-conserving networks); (ii) symbolic recovery modules that explicitly extract physical constants rather than approximating them; (iii) training objectives derived internally from safety-critical evaluation constraints rather than soft penalty formulations.

10 Conclusion

We presented ThermoPINN — a physics-constrained meta-learning system for turbofan engine RUL prediction — and used it to conduct a large-scale empirical evaluation of whether soft-constrained PINNs recover governing physical laws under distribution shift.

The results provide strong empirical evidence that they do not.

Despite outperforming published baselines on both synthetic (RMSE = 40.10 at 10-shot) and real-world (RMSE = 31.88, zero-shot N-CMAPSS) benchmarks, and despite satisfying all four internal safety and calibration criteria, all tested physics laws fail hypothesis testing ($p < 0.05$). The Paris–Erdogan exponent deviates by 57%, the Arrhenius activation energy by 60%, and the Norton–Bailey stress exponent by 55%.

Across distribution shifts, epistemic uncertainty fails to increase proportionally and instead exhibits collapse behavior, while survival-function ranking degrades toward near-random performance.

These observations are not attributable solely to implementation or optimization choices. Rather, they are consistent with structural properties of the soft-constrained PINN formulation, as captured by the PINN Misalignment Principle ($\min_{\theta} \mathcal{L}_{\text{train}}(f_{\theta}) \not\Rightarrow f_{\theta} \in \mathcal{F}_{\text{phys}}$, Proposition 1)

and mechanistically explained by the Manifold Projection Hypothesis.

The implication for aerospace machine learning is that soft-constrained PINNs may not be sufficient to guarantee physically consistent internal representations under distribution shift, even when predictive accuracy is high. This suggests a gap between predictive performance and physical validity that must be explicitly addressed in safety-critical settings.

11 Reproducibility Statement

All experiments in this work are fully reproducible under the configurations described below.

- **Code:** The complete implementation, including architecture definitions for ThermoPINN and baselines, is available at: <https://github.com/GURU1001S/ThermoPINN>
- **Dataset:** The UTDTB v5 synthetic benchmark is archived on Zenodo at DOI: [10.5281/zenodo.19654819](https://doi.org/10.5281/zenodo.19654819). This HDF5 file includes all 55-dimensional feature streams and causal physics labels.
 - **Real-world benchmarks:** Evaluation on N-CMAPSS (DS01–DS05) utilized the official NASA repository. For reproducibility, the processed HDF5 versions are available via: https://github.com/mohyunho/N-CMAPSS_DL
- **Model Checkpoint:** The exact weights (v19.pt) used to generate the results in Section 7 are provided in the GitHub repository for immediate verification.
- **Computation:** Experiments were conducted on an NVIDIA RTX 3050 (6 GB VRAM). While ThermoPINN training is highly efficient, zero-shot transfer inference across the full 26 million N-CMAPSS evaluation windows requires approximately 4.5 hours of processing time.
- **Preprocessing:** All trajectories undergo per-engine Z-score normalisation. No data leakage occurs between the training sets of UTDTB v5 and the OOD N-CMAPSS evaluation sets.

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A Complete Model Configuration

Table 7: ThermoPINN final configuration (v19).

Parameter	Value
Total parameters	2,992,892
Encoder conv channels	256
GRU hidden	512
Head hidden	128
Dropout	0.30
Transformer layers	2
Transformer heads	8
Feed-forward dim	512
Inner steps K	2
Inner LR α	10^{-3}
Eval inner LR factor	0.25
Encoder LR	3×10^{-5}
Head LR	1.28×10^{-4}
Gradient clip norm	1.0
Tasks per batch M	12
Accumulation steps	4
Batch size	128
Training epochs	300
Cosine $T_{\max}$	300
Cosine $\eta_{\min}$	10^{-6}
film.alpha init	-3.0
degradation head bias	-3.0

B Physics Loss Weights

Table 8: Physics loss λ schedule (v19).

Term	λ
Asymmetric Huber (data)	1.00
Monotonic degradation	0.10
Health BCE	0.20
Uncertainty NLL	0.10
Physics direct supervision	0.50
Inner damage monotonicity	0.50
Burst risk BCE	0.20

C NLS Regression Procedure

For each governing law $\mathcal{G}(y; \theta)$ with parameters θ , we:

1. Extract the relevant latent node outputs from ThermoPINN on 1,000 held-out test trajectories.
2. Initialise NLS at the theoretical value $\theta_0 = \theta_{\text{true}}$ (Levenberg–Marquardt, 500 iterations).
3. Test $H_0 : \theta = \theta_{\text{true}}$ via KS test on the distribution of residuals under the theoretical vs. fitted model.
4. Report fitted $\hat{\theta}$, relative error $|\hat{\theta} - \theta_{\text{true}}|/\theta_{\text{true}}$, and p -value.

All regressions are initialised at θ_{true} to eliminate initialisation bias. Results are robust to 10 different random restarts within $\pm 50\%$ of θ_{true} .

D Sensor Robustness Paradox

Under increasing sensor noise and dropout, ThermoPINN RMSE *improves* rather than degrades:

Table 9: RMSE under sensor degradation (100 test engines). The baseline condition here uses the full 55-D feature vector including latent physics channels; this differs from the zero-shot evaluation in Table 1 (RMSE = 92.99), which uses only the 36 observable features available at inference time on external datasets.

Condition	RMSE	Δ
No noise	100.0	baseline
Moderate noise	92.0	-8.0%
Severe noise	89.0	-11.0%
15% sensor dropout	91.6	-8.4%

This paradox arises because sensor noise partially masks spurious correlations in the raw sensor stream, forcing the model to rely more heavily on the 19 latent physics state channels — which carry the true degradation signal with higher signal-to-noise ratio.

This finding supports the UTDTB v5 design hypothesis that causal physics states are more informative than raw telemetry for predicting RUL.

References

- [1] Manuel Arias Chao, Chetan Kulkarni, Kai Goebel, and Olga Fink. Aircraft engine run-to-failure dataset under real flight conditions for prognostics and diagnostics. In *Data*, volume 6, page 5. MDPI, 2021.
- [2] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. In *arXiv preprint arXiv:1803.01271*, 2018.
- [3] Cannon-Muskegon Corporation. Cmsx-4 superalloy material properties, 2000. Nickel-based single crystal superalloy data sheet.
- [4] Salvatore Cuomo, Vincenzo Schiano Di Cola, Fabio Giampaolo, Gianluigi Rozza, Maziar Raissi, and Francesco Piccialli. Scientific machine learning through physics-informed neural networks: where fractional calculus meets artificial intelligence. *Journal of Scientific Computing*, 92(3):88, 2022.

- [5] Antonio Dourado and Felipe Viana. Physics-informed neural networks for prognostics and health management. *Mechanical Systems and Signal Processing*, 2021.
- [6] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning*, pages 1126–1135. PMLR, 2017.
- [7] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation. *ICML*, 2016.
- [8] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2021.
- [9] Ehsan Haghighat and Ruben Juanes. A physics-informed deep learning framework for structural mechanics. *Computer Methods in Applied Mechanics and Engineering*, 2021.
- [10] George Em Karniadakis, Ioannis G Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3(6):422–440, 2021.
- [11] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, et al. WILDS: A benchmark of in-the-wild distribution shifts. In *International Conference on Machine Learning*, pages 5637–5664. PMLR, 2021.
- [12] Aditi Krishnapriyan, Amir Gholami, Shandian Zhe, Robert Kirby, and Michael W Mahoney. Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 26548–26560, 2021.
- [13] X. Li et al. Physics-informed neural networks for remaining useful life prediction. *Reliability Engineering & System Safety*, 2022.
- [14] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.
- [15] X. Ma et al. Meta-learning for domain adaptation in rul prediction. *IEEE Transactions on Industrial Informatics*, 2022.
- [16] Takashi Matsubara, Ai Ishikawa, and Takaharu Yaguchi. Physics-embedded neural networks: Graph neural PDE solvers with mixed boundary conditions. In *Advances in Neural Information Processing Systems*, volume 35, pages 23218–23229, 2022.
- [17] Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, Dilan Gorur, and Balaji Lakshminarayanan. Do deep generative models know what they don’t know? In *International Conference on Learning Representations*, 2019.
- [18] Yaniv Ovadia et al. Can you trust your model’s uncertainty? *NeurIPS*, 2019.
- [19] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. 32, 2018.
- [20] Aniruddh Raghu, Maithra Raghu, Samy Bengio, and Oriol Vinyals. Rapid learning or feature reuse? towards understanding the effectiveness of MAML. In *International Conference on Learning Representations*, 2020.
- [21] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [22] Abhinav Saxena and Kai Goebel. Turbofan engine degradation simulation data set. 2008. NASA Ames Prognostics Data Repository.
- [23] Abhinav Saxena, Kai Goebel, Don Simon, and Neil Eklund. Damage propagation modeling for aircraft engine run-to-failure simulation. pages 1–9, 2008.
- [24] Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [25] Albert Tarantola. *Inverse Problem Theory and Methods for Model Parameter Estimation*. SIAM, 2005.
- [26] Sifan Wang, Yujun Teng, and Paris Perdikaris. Understanding and mitigating gradient flow pathologies in physics-informed neural networks. *SIAM Journal on Scientific Computing*, 43(5):A3055–A3081, 2021.
- [27] L. Wen et al. Transformer-based rul prediction for turbofan engines. In *IEEE*, 2022.
- [28] Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long. TimesNet: Temporal 2D-variation modeling for general time series analysis. In *International Conference on Learning Representations*, 2023.

- [29] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023.
- [30] Shuang Zheng et al. Long short-term memory network for remaining useful life prediction. In *IEEE PHM Conference*, 2017.